Inverse Inference of D0L Systems: an Exact Algorithm, Bounds, and the Binary Threshold

Abstract

Given a word w and a number of steps n, we ask for a D0L system (an axiom and context-free rewriting rules) that derives w in exactly n steps. We give a pruned backtracking search and prove it sound and complete against a brute-force specification; we prove existence theorems, a complete characterization over a one-letter alphabet, two pigeonhole bounds showing that most words have no small system; and, for a two-letter alphabet, a threshold $T(n) = \min(F_{n+1}, D(n))$ below which every non-erasing solution with a short axiom is an *insertion system*, together with a proof that $T(n)$ is attained for every $n \geq 2$. Every statement is machine-checked in Lean 4 (core library only, no `sorry`), and the statements in this paper are aligned to the formalization with PFA tags.

1 Definitions

Throughout, an alphabet is a finite list Σ of symbols; for the general results we take symbols in $\mathbb{N}$, and for the binary results the two symbols are the booleans $a = \mathsf{false}$, $b = \mathsf{true}$.

Definition 1 (D0L system)**.** A *D0L system* over Σ is a pair $S = (\omega, f)$ of an *axiom* $\omega \in \Sigma^*$ (formally `start`) and a *rule* map $f : \Sigma \to \Sigma^*$.

Definition 2 (Derivation)**.** One *step* applies f to every letter and concatenates the results. The word *derived in n steps* is $S.\mathrm{derive}(n) = f^n(\omega)$, with $S.\mathrm{derive}(0) = \omega$.

Example 3. For the Fibonacci system $\mathrm{fib}_{\mathbb{N}}$: $0 \mapsto 01$, $1 \mapsto 0$ with axiom 0, five steps give 0100101001001.

Definition 4 (Bounds, presented systems, the solution set)**.** A bound B consists of a maximal axiom length, a maximal rule length, and whether empty rules are allowed. A *presented* system is an axiom together with an association list of rules; `allSystems` (Σ, B) enumerates every presented system within B. The *solution set* $\mathrm{sol}(\Sigma, B, w, n)$ is the list of presented systems in `allSystems` that derive w in n steps.

Definition 5 (The search)**.** `infer` $(\Sigma, B, w, n, \mathrm{fuel})$ is a depth-first search that builds the derivation tree of w leftmost-first, inventing rule symbols one at a time and sharing each rule across all of its occurrences; it prunes a branch as soon as the leaves still owed by the stack cannot fit in the remainder of w. The `fuel` argument bounds recursion depth.

Definition 6 (Canonical form)**.** Two presented systems are identified by canon when they have the same axiom and agree on every symbol that is actually rewritten during the first n steps.

2 Existence

Theorem 7 (Trivial existence)**.** *For every word w and every n, there is a D0L system S with* $S.\mathrm{derive}(n) = w$.

Proof. Take axiom w and the identity rules. □

Theorem 8 (One hidden symbol)**.** *For every word w and every n with $n \geq 1$, there is a system over $\Sigma \cup \{\#\}$ with an axiom of length 1 deriving w in n steps.*

Proof. Axiom #, rule $\# \mapsto w$, identity on Σ. □

Theorem 9 (Chain)**.** *Let w be a word and $c_0, \ldots, c_{n-1}$ a list of distinct letters, each occurring in w, of length n with $n \geq 1$. Then there is a system with a one-letter axiom, deriving w in n steps, whose axiom letter lies in w and whose rules are the identity outside the letters of w.*

Proof. $c_0 \mapsto c_1 \mapsto \cdots \mapsto c_{n-1} \mapsto w$. □

3 The search is exact

Theorem 10 (Completeness)**.** *Let Σ be an alphabet, B a bound, w a word, n a step count and P a presented system with $P \in \mathrm{sol}(\Sigma, B, w, n)$. Then for some fuel, the search emits a system Q with* $\mathrm{canon}(Q) = \mathrm{canon}(P)$.

Theorem 11 (Soundness)**.** *Let Σ, B, w, n be as above. If Q is emitted by the search with any fuel, $Q \in \mathit{infer}(\Sigma, B, w, n, \mathrm{fuel})$, then there is $P \in \mathrm{sol}(\Sigma, B, w, n)$ with* $\mathrm{canon}(Q) = \mathrm{canon}(P)$.

Proof sketch. Completeness: fix a solution, lift it to a system over $\Sigma \cup \{\#\}$ whose #-rule is the axiom, and show by induction on fuel that the search state stays *on the path* of that system—the leaves still owed by the stack are exactly the unread suffix of w, closed rules are exact and open rules are prefixes—while a measure $3 \cdot \mathrm{pending} + 2 \cdot |\mathrm{stack}| + \mathrm{leaves} + \mathrm{slack}$ strictly decreases. The pruning test is a lower bound on the owed leaves, so it never fires on the path. Soundness is the mirror image: induct backwards from the emitted key, read a system off the final rule table, and carry an extension relation between the current and final tables. Three details of the search are what make soundness unconditional: leaf candidates are restricted to Σ, an empty axiom is never closed (empty rules may be), and the search returns nothing when no legal rule body exists; without them the search and the brute-force set differ on $w \notin \Sigma^*$, on $w = \varepsilon$, and on a zero rule-length cap. □

4 Impossibility and counting

Theorem 12 (Unary alphabet)**.** *For L, n with $L \geq 1$ and $n \geq 1$, the word a^L is derived in n steps by a one-letter system with an axiom shorter than L if and only if $m^n \mid L$ for some $m \geq 2$.*

Proof. $|f^n(a^j)| = j\,|f(a)|^n$; $|f(a)| \in \{0, 1\}$ forces $L = 0$ or $j = L$. □

Theorem 13 (Counting under per-rule caps)**.** *Let Σ be an alphabet without repetitions, B a bound and n, L naturals with fewer bounded systems than $|\Sigma|^L$. Then some word of length L over Σ has no bounded system at step n.*

Theorem 14 (Incompressibility)**.** *Let Σ be without repetitions, with $k = |\Sigma|$, and n, L, s naturals with $(k+1)^{s+k+1} < k^L$. Then some word of length L over Σ is not derived in n steps by any system over Σ (identity outside Σ) whose description size $|\omega| + \sum_{a\in\Sigma} |f(a)|$ is at most s.*

Proof. A system of size $\leq s$ is encoded as $\omega \# f(a_1) \# \cdots \# f(a_k) \# \cdots \#$, padded to length $s+k+1$ over $\Sigma \cup \{\#\}$; decoding splits on $\#$. Pigeonhole on lists: a duplicate-free list contained in another is no longer. □

5 The binary threshold

For the rest of the section, systems are over $\{a, b\}$, *non-erasing* ($f(c) \neq \varepsilon$), and the axiom is strictly shorter than the derived word, which uses both letters. An *insertion system* fixes one letter x and sends the other to $x^p\, \bar{x}\, x^q$ with $p + q \geq 1$. Write F_n for the Fibonacci numbers ($F_0 = 0$, $F_1 = 1$) and $D(n) = 2 \cdot 2^{\lfloor n/2 \rfloor}$ for even n, $3 \cdot 2^{\lfloor n/2 \rfloor}$ for odd n.

Definition 15 (Threshold)**.** $T(n) = \min(F_{n+1}, D(n))$.

Example 16. $T(0), \ldots, T(9) = 1, 1, 2, 3, 5, 8, 13, 21, 32, 48$.

Theorem 17 (Sharp bound)**.** *Let S be non-erasing and n, w with $S.\mathrm{derive}(n) = w$, $|\omega| < |w|$, w containing both letters and $|w| < T(n)$. Then S is an insertion system.*

Example 18. The hypotheses are satisfiable: $b \mapsto ab$, $a \mapsto a$, axiom b, $n = 5$, $w = a^5 b$.

Proof sketch. If both rules have length ≥ 2 the length doubles each step. Otherwise some letter x has $|f(x)| = 1$. If $f(x) = x$, the number of $\bar{x}$'s is multiplied by $\beta = |f(\bar{x})|_{\bar{x}}$ each step: $\beta \geq 2$ gives $|w| \geq 2^n$, $\beta = 0$ erases $\bar{x}$, so $\beta = 1$ and $f(\bar{x}) = x^p \bar{x} x^q$. If $f(x) = \bar{x}$, the Parikh vector (p, q) of $(x, \bar{x})$ obeys $p' = \alpha q$, $q' = p + \beta q$: for $\alpha, \beta \geq 1$ it dominates the Fibonacci recurrence and $|w| \geq F_{n+1}$; for $\beta = 0$ it scales by $\alpha \geq 2$ every two steps and $|w| \geq D(n)$; $\alpha = 0$ makes w unary. □

Insertion words have an explicit description. For $x \in \{a, b\}$ let $\mathrm{runs}_x(w)$ be the list of lengths of the maximal x-blocks of w, separated by the occurrences of $\bar{x}$.

Definition 19 (Run-length criterion)**.** $\mathrm{Crit}(n, x, w)$ holds when w contains $\bar{x}$, every *inner* run of $\mathrm{runs}_x(w)$ has length $\geq n$, and the leading or the trailing run has length $\geq n$.

Theorem 20 (Insertion words satisfy the criterion)**.** *Let S fix the letter x, $f(x) = x$, and send the other to $f(\bar{x}) = x^p \bar{x} x^q$ with p, q, n and $p + q \geq 1$. If $S.\mathrm{derive}(n) = w$ and $\bar{x}$ occurs in w, then $\mathrm{Crit}(n, x, w)$.*

Theorem 21 (Words meeting the criterion are insertion words)**.** *Let n with $n \geq 1$, x a letter and w a word with $\mathrm{Crit}(n, x, w)$ whose leading x-run has length $\geq n$. Then some system fixing x, with axiom shorter than w, derives w in n steps.*

Proof. Remove n copies of x before every $\bar{x}$ to form the axiom; the rules $x \mapsto x$, $\bar{x} \mapsto x\bar{x}$ put them back. The trailing-run case is the reversal. □

Theorem 22 (Sharpness)**.** *For every n with $n \geq 2$, there is a non-erasing system with an axiom shorter than the word it derives in n steps, that word using both letters, of length exactly $T(n)$, and satisfying neither $\mathrm{Crit}(n, a, \cdot)$ nor $\mathrm{Crit}(n, b, \cdot)$.*

Proof. For $n \le 7$ take $a \mapsto ab$, $b \mapsto a$ from axiom b: the iterates have length F_{n+1}, contain no bb and no aaa (an invariant preserved by the step), while the criterion forces a factor x^n. For $n \ge 8$ take $a \mapsto b$, $b \mapsto aa$ from axiom ab: the iterates are $a^{2^k}b^{2^k}$ or $b^{2^k}a^{2^{k+1}}$, of length $D(n)$, and a two-block word has an inner run of length 0. Finally $D(n) \le F_{n+1}$ exactly when $n \ge 8$, since $D(n+2) = 2D(n)$ and $F_{n+3} \ge 2F_{n+1}$. □

Consequently, below $T(n)$ a binary word is derivable (non-erasing, short axiom) exactly when it satisfies $\mathrm{Crit}(n, a, \cdot)$ or $\mathrm{Crit}(n, b, \cdot)$, and the bound $T(n)$ is attained for every $n \ge 2$.

6 The open region

Above $T(n)$ the question is decidable (by the search) but no structural description is known. Exhaustive search for $n \le 6$ and $|w| \le 15$ shows three families: insertion, Fibonacci-type ($a \mapsto ab$, $b \mapsto a$ and relatives, growth φ^n), and doubling ($a \mapsto b$, $b \mapsto aa$, growth $\sqrt{2}^n$), together with longer-axiom mixtures such as $a^4b^4a^4$ at $n = 4$ from $a \mapsto bb$, $b \mapsto a$, axiom aba.

Conjecture (Density). For fixed $n \ge 3$ let $N_n(L)$ count the binary words of length L with a non-erasing short-axiom system at step n. Then $N_n(L)^{1/L} \to \rho_n$, the dominant root of $x^{n+1} = x^n + 1$, i.e. the growth rate of the insertion family. Every family with both letters active expands its axiom by a factor $\ge 2^{\lfloor n/2 \rfloor}$ and so has growth rate at most $2^{2^{-\lfloor n/2 \rfloor}}$, exponentially closer to 1.

Formalization

All definitions and theorems above are tagged to `Paper.lean` (Lean 4.36.0-rc2, core library, no `sorry`); the tagged statements were checked against the formal signatures with the PFA checker. The exact algorithm and all auxiliary lemmas live in the `LSys` namespace; the paper-facing statements in `Paper` fix the alphabet to $\mathbb{N}$, `Unit` or `Bool` and are proved by appeal to the general ones.